

Project Design Phase

Solution Architecture

Date	8 September 2026
Team ID	Individual Project
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Project Name	An Employee Requests Installation of Licensed Software through the Service Catalog
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture defines how the ServiceNow components work together to automate the complete lifecycle of a licensed software installation request. It bridges the business problem of manual and unstructured software requests with a centralized, automated, and trackable ServiceNow solution.

- Provide a standardized Service Catalog interface for employees to request licensed software.
- Automate request processing using ServiceNow Flow Designer.
- Support approval, notification, task creation, and fulfillment in a structured sequence.
- Maintain request information in ServiceNow tables for centralized visibility and tracking.
- Ensure the request can be managed from submission through fulfillment and closure.

Architecture Components:

- 1. Employee / Requester:** The employee accesses the ServiceNow portal and submits a request for installation of licensed software.
- 2. Service Catalog:** A Software Installation Request catalog item collects details such as Software Name, Version, License Justification, and Urgency.
- 3. ServiceNow Request Tables:** The submitted request is stored and tracked using ServiceNow request-related records, including `sc_request`, `sc_req_item`, and `sc_task`.
- 4. Flow Designer:** The Catalog Item Requested flow automates the business process after the employee submits the catalog request.
- 5. Email Notification:** A Send Email action sends a confirmation notification to the requester after the request is submitted.
- 6. Approval Process:** The Ask for Approval action routes the request through the required approval process before fulfillment.
- 7. Catalog Task:** After the approval stage, a catalog task is created and assigned to the Software Support Team for software installation.
- 8. IT Support Team:** The assigned support team performs the fulfillment activity and updates the task status.

9. Status Tracking and Closure: The requester and support team can track the request through ServiceNow until fulfillment is completed.

Solution Architecture Diagram:

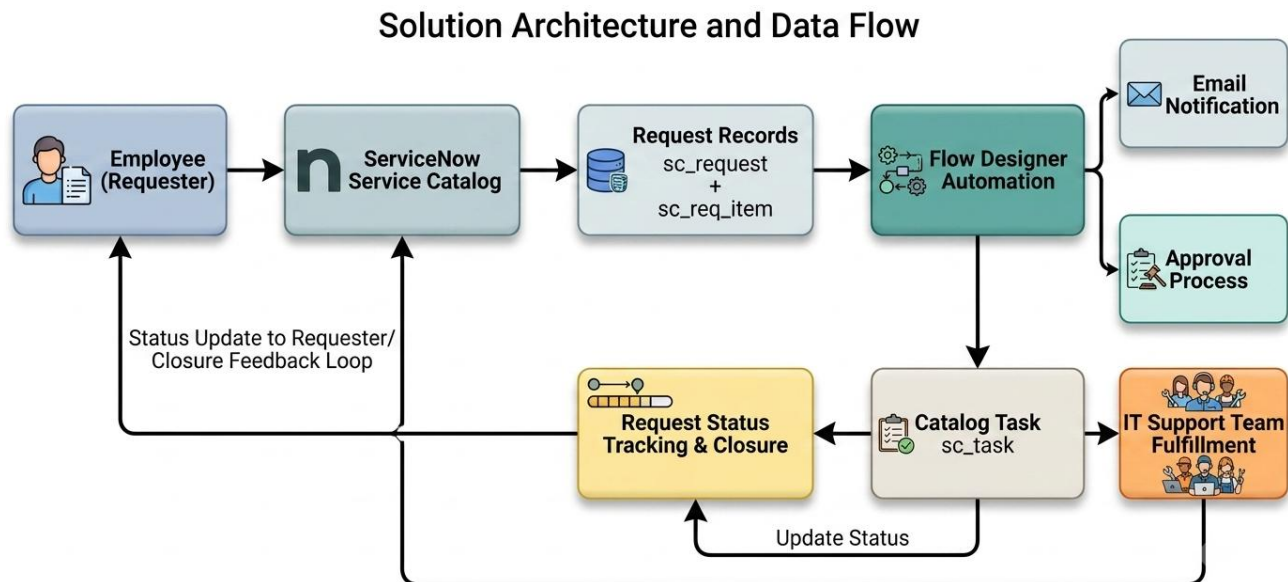


Figure 1: Architecture and data flow of the Software Installation Request Automation solution

Data Flow Explanation:

1. The employee opens the ServiceNow Service Catalog and selects the Software Installation Request item.
2. The employee enters the required software request details and submits the request.
3. ServiceNow creates the corresponding request and requested item records.
4. The Flow Designer is triggered when the catalog item is requested.
5. A confirmation email is sent to the requester.
6. The request enters the approval stage.
7. A catalog task is created for the Software Support Team for fulfillment.
8. The support team processes the task and updates the request/task status.
9. The request lifecycle is tracked in ServiceNow until completion.

Architecture Benefits:

- Reduces manual coordination through email and informal communication.
- Improves approval and fulfillment efficiency.
- Provides centralized visibility of software requests.
- Improves accountability through task assignment to the Software Support Team.
- Supports compliance by standardizing licensed software requests.
- Enhances employee experience through transparent request tracking.